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## **A Comparative Analysis of Classical Approaches vs. Machine Learning Methodologies for Property Modeling Using EMBER**

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### **Abstract**

This study evaluates the effectiveness of the Embedded Model Estimator (EMBER), a machine learning-based tool, designed to improve porosity modeling within petrophysical workflows. EMBER's performance is tested in two field development plan (FDP) projects: a clastic reservoir in the Sirte Basin and a carbonate reservoir in the Intisar Field, both located in Libya. Due to their significant geological heterogeneity, these reservoirs provide a good example for comparing EMBER with traditional geostatistical modeling techniques.

In addition to the porosity in this study we incorporate facies models and stratigraphic seismic attributes, including acoustic impedance, into the EMBER workflow to have better representation for the porosity distribution.

The clastic reservoir lies within a structurally complex area between the Zaltan and Al Jahamag platforms, featuring faulted anticlines and a stratigraphic succession comprising rocks spanning from the Cambro-Ordovician Gargaf to the Upper Cretaceous Bahi Formations. In contrast, the carbonate reservoir in the Intisar Field comprises pinnacle reef structures belonging to the Zaltan carbonate shelf, characterized by hydrocarbon accumulation in the highly heterogeneous Upper Sabil limestones. Results demonstrate that the EMBER algorithm consistently produces porosity models that closely align with well log measurements while preserving geological continuity. In the carbonate reservoir, the EMBER algorithm slightly outperforms classical methods in terms of accuracy and distribution fidelity, while in the clastic reservoir, EMBER outperforms the classical methods. Across both formations, the method significantly reduces modeling time - by over 70% - without compromising data integrity. This time efficiency, coupled with its high predictive performance, highlights the value of the EMBER's algorithm especially in complex geological settings.

Unlike conventional approaches that rely on fixed statistical assumptions, EMBER uses machine learning algorithms to learn adaptive relationships between geological and geophysical inputs. This allows more accurate and realistic modeling of subsurface heterogeneity. The study proves EMBER's advantages in enhancing modeling accuracy, operational efficiency, and reducing geological uncertainty, reinforcing its value in hydrocarbon reservoir characterization and development planning.

## Introduction

Porosity modeling in geologically complex reservoirs remains a challenging and time-consuming component of reservoir characterization. Traditional geostatistical methods, widely used in Field Development Planning (FDP), require extensive expertise in data conditioning, variogram modeling, and algorithm configuration. While these classical techniques are robust, their workflows can be tedious and highly dependent on user experience and statistical knowledge.

To overcome these limitations, the Embedded Model Estimator (EMBER) presents a modern, machine learning-based alternative. EMBER automatically learns the relationships between geological and geophysical inputs, eliminating much of the manual effort associated with traditional approaches. It significantly reduces modeling time while delivering porosity models with accuracy comparable to - or better than - those produced by geostatistical methods.

This study evaluates EMBER's performance against conventional techniques across two FDP projects located in Libya's Sirte Basin - one of North Africa's most prolific hydrocarbon provinces. Porosity models generated by EMBER and traditional geostatistical workflows are compared here in terms of both prediction accuracy and computational efficiency. Through this comparison, the paper demonstrates EMBER's practical advantages in streamlining complex reservoir modeling while maintaining geologically consistent results.

## Geological setting

The Sirte Basin is a Mesozoic–Cenozoic rift basin formed by extensional tectonics along the southern margin of the Tethys Ocean. Characterized by horst and graben structures and thick sedimentary sequences, it hosts several petroleum systems sourced from the Upper Cretaceous Sirte Shale, Rachmat Formation and Nubian sandstones (Ahlbrandt, 2001; Burwood et al., 2003; El-Alami et al., 1989; Belazi, 1989; Abdunaser, 2015). Vertical hydrocarbon migration through fault zones and trapping beneath Eocene evaporites has led to significant accumulations across multiple reservoir intervals.

Three main reservoirs are investigated in this study.

The Bahi Formation represent Upper Cretaceous units deposited over older Paleozoic and Mesozoic rocks on structural highs in the eastern Sirte Basin. Composed of fine-grained sandstones and poorly sorted conglomerates interbedded with claystones, these formations exhibit highly variable reservoir quality, from poor, in conglomeratic facies, to excellent in well-sorted marine sandstones, with porosity up to 30% and permeability reaching 1000 mD.

The Qarqaf Group (Gergaf Sandstone) comprises Cambrian–Ordovician quartzose sandstones deposited in fluvial to shallow marine environments. Although typically tight (low porosity and permeability), with localized increase in reservoir properties, related to fracturing or favorable diagenesis (cementation, compaction). the Qarqaf Group sediments are known to generally underlie a Mesozoic and Tertiary overburden in much of the central part of the Sirte Basin. Qarqaf Group reservoirs are noted at the northern end of the Zaltan Platform (e.g. in the Attahadi and Hateibi gas condensate fields), where they are sourced by the prolific Sirte Shale in the Agedabia Trough. Environments are interpreted to represent fluvial braided channels, marginal marine estuarine/deltaic channels/bars and shallow marine shoals. Tidal facies are locally noted in some study wells. The source of these quartz-rich sandstones is thought to be the Pan African Mountains to the south. Deposition took place on an unstable continental margin of Pannotia and subsequently Gondwana during the Cambrian–Ordovician (Hallett, 2002).

The Upper Sabil Reservoir, in the Intisar D Field, represents a high-quality Paleocene-aged carbonate build-up deposited on a paleotopographic high within a shallow marine environment. This reservoir is locally referred to as the "Pencil Reef" due to its narrow, vertically elongated geometry, characteristic of isolated carbonate bodies formed in platform margin to ramp settings. In terms of geometry, the Sabil build-up is localized and vertically stacked, making it an excellent stratigraphic trap for hydrocarbons. The

porosity values within the Upper Sabil commonly range from 25% to 35%, with permeability influenced by both original depositional texture and diagenetic overprint. The sedimentological study concluded that the reservoir quality of the Upper-Sabil and Harash deposits inside the Intisar-D build-up is related to the presence of intergranular, intragranular, mouldic and vuggy porosity. The key factors controlling the reservoir quality are the original depositional texture (the amount of fine matrix) and the diagenetic modifications affecting these sediments: the dissolution of grains that enhanced the reservoir quality, and the occlusion of pores by calcite and/or dolomite cements that reduced the reservoir quality.

Overall, the Upper Sabil Reservoir reflects a classic example of a carbonate build-up system formed in a shallow marine, high-energy environment, later enhanced by diagenetic processes which ones?, making it the principal hydrocarbon-bearing unit.

## Methodology

Porosity modeling is a fundamental component of static reservoir characterization, directly influencing volumetric calculations and reservoir performance predictions. Classical geostatistical approaches—such as Sequential Gaussian Simulation (SGS)—rely on statistical interpolation constrained by spatial continuity, typically quantified through variogram modeling. These methods assume stationarity and require careful data transformation (e.g. normal score transformation) to ensure Gaussian behavior, which is essential for most geostatistical algorithms.

In contrast, the Embedded Model Estimator (EMBER) adopts a machine learning framework to infer relationships between geological and geophysical variables without relying on strict statistical assumptions. EMBER incorporates supervised learning techniques within the geostatistical workflow, using secondary variables (e.g., acoustic impedance) as soft conditioning data. It dynamically adjusts to spatial patterns present in the training dataset, offering a more flexible and user-friendly alternative to traditional methods.

In this study, two modeling approaches—classical SGS and EMBER—were applied to two distinct reservoir models. Both workflows were implemented in Petrel. Input data included wireline log-derived porosity, interpreted facies models, and acoustic impedance volumes. Variogram analysis and histogram conditioning were used for the classical method, whilst for the EMBER workflow, machine learning training was performed on well and seismic data, with no need for manual variogram modeling nor transformation to done.

For the clastic reservoir model, 53 drilled wells were distributed across the study area. while, the 103D reservoir model included 16 wells, with their distribution concentrated around the reef buildup.

Reservoirs in the Sirte Basin exhibit high degrees of lateral heterogeneity, fault compartmentalization, and mixed lithofacies, which challenge the assumptions of stationarity and Gaussian distribution. The EMBER's data-driven, adaptive, nature is particularly suited for such complex geological environments, providing robust models where classical approaches may struggle.

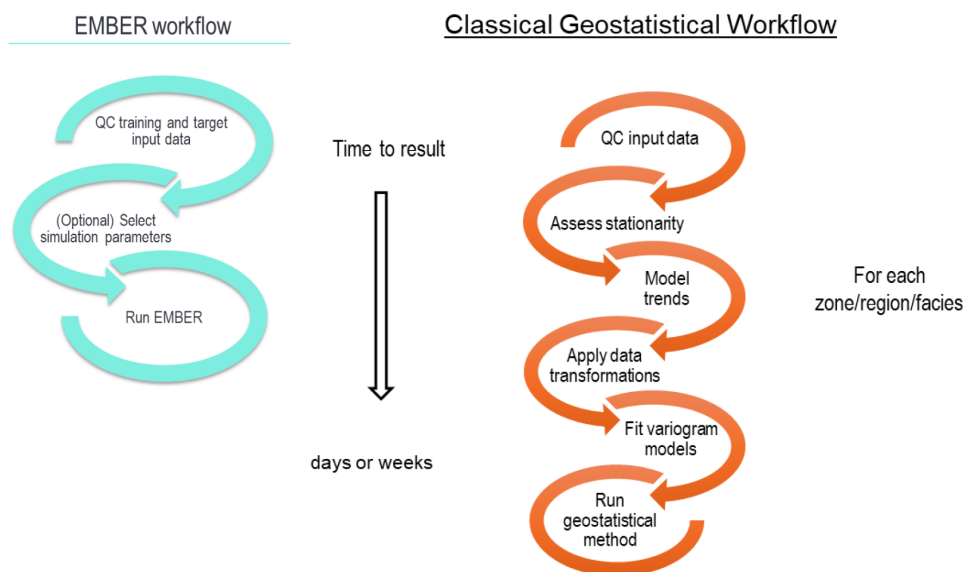


Figure 1—Ember workflow vs classic geostatistical methods workflow

## Workflow

1. **Data Collection:** Data was collected from various sources, including well logs (e.g., porosity logs), seismic data (such as amplitude and impedance attributes), and geological interpretations (like facies models).
2. **Model Training:** Machine learning–based property modeling allows the use of an ensemble of tree-based estimators. By combining multiple decision trees, it is possible to generate high-quality predictive models. EMBER enhances this process by transforming geostatistical model predictions into input features, which are then used—alongside other continuous and categorical training features—to train the machine learning algorithm.
3. **Generate simulation model:** Once the training target and features have been defined for the machine learning engine, you can proceed to the Simulation tab to specify the parameters for generating a (stochastic) simulation. The trained model produces a robust set of conditional distributions at each grid cell, from which stochastic realizations can be drawn. These simulations introduce the necessary variability and heterogeneity into the model, which are essential for accurately constraining flow simulations. The resulting characteristics can be applied either globally or locally across the grid.
4. **Validation and Blind Testing:** To ensure accuracy, the model is validated using methods like cross-validation or blind well tests, where known data from a well is temporarily hidden and then predicted by the model. The predicted values are compared to the actual values to measure performance.
5. **Result Interpretation and Refinement:** The output porosity model was visualized, analyzed and compared with traditional geostatistical models. In many cases, EMBER’s results were as accurate and often more geologically consistent than the classical modeling approach, especially in data-rich environments.

## Presentation of Data and Results

### Reservoir Modeling Approach and Results: Bahi and Gergaf (Classical Method)

All wireline logs capturing petrophysical properties, like porosity or volume of shale, were estimated. In the Bahi and Gergaf reservoirs,  $V_{sh}$  is calculated from wireline log (Gamma ray, Neutron and Density logs). The  $V_{sh}$  log was upscaled arithmetically. The property was distributed using Sequential Gaussian

Simulation (SGS) algorithm within petrel. Vsh was distributed within each zone separately, using zone specific distribution parameters.

The primary guide for Vsh distribution was the data transformations and variograms (from petrel data analysis). The trend maps for Vsh (as explained in the previous section) were used as a secondary control for guiding the lateral extent of Vsh distribution. Comparison of input Vsh histogram and static model demonstrates a reasonable match (Figure Below).

Porosity was up-scaled arithmetically from the porosity log. Porosity distribution was performed in the 3-D geocellular model using SGS algorithm, with zone specific distribution.

Primary guides for the modeling algorithm were data transformations and variogram (from data analysis), whereas, Vsh was used as a secondary control. Porosity distributions by zone were edited within data analysis

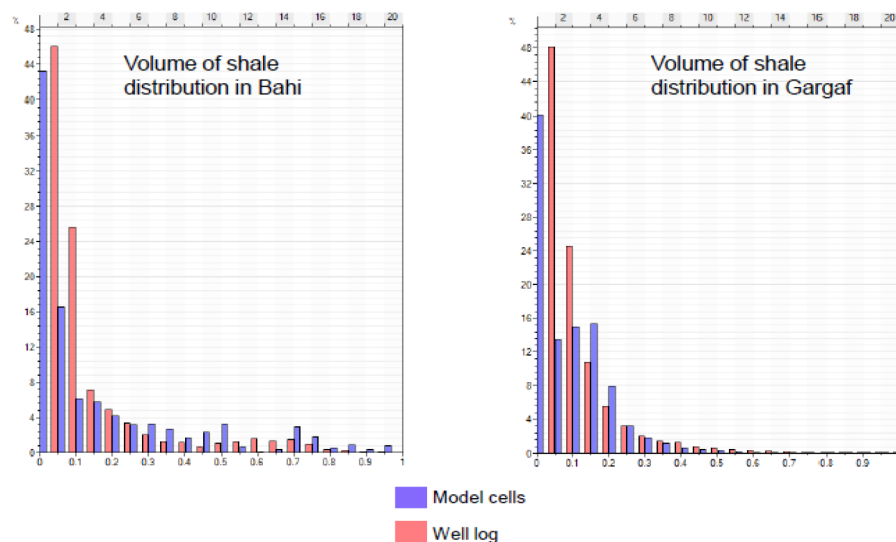


Figure 2—Volume of shale from the model vs input data

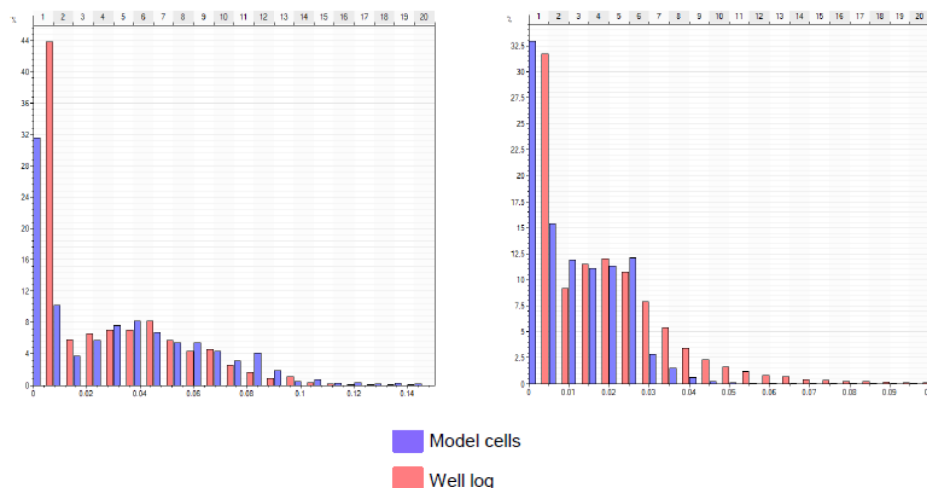


Figure 3—Porosity model vs input data (logs)

### Reservoir Modeling Approach and Results: Bahi and Gergaf (EMBER Method)

The input was the upscaled porosity and the Vsh which was used as secondary data during the previous workflow; no additional adjustments to the defaults were used when distributing the Porosity using Ember.

Ember shows better predictabilities in most of the intervals, however, the classical method shows better prediction in others. nevertheless, and overall, Ember shows better prediction when a blind well test was applied.

The predicted porosity log comparison indicates that Ember outperforms the classical method in estimating porosity at the well location. Additionally, Ember provides a more accurate representation of porosity in relation to the volume of shale. The figure below illustrates the comparison between upscaled porosity and predictions from both the EMBER and classical geostatistical methods, plotted alongside the volume of shale (Vsh). The porosity predicted by EMBER shows a stronger alignment with the Vsh trends, indicating its improved ability to capture the relationship between shale content and porosity distribution.

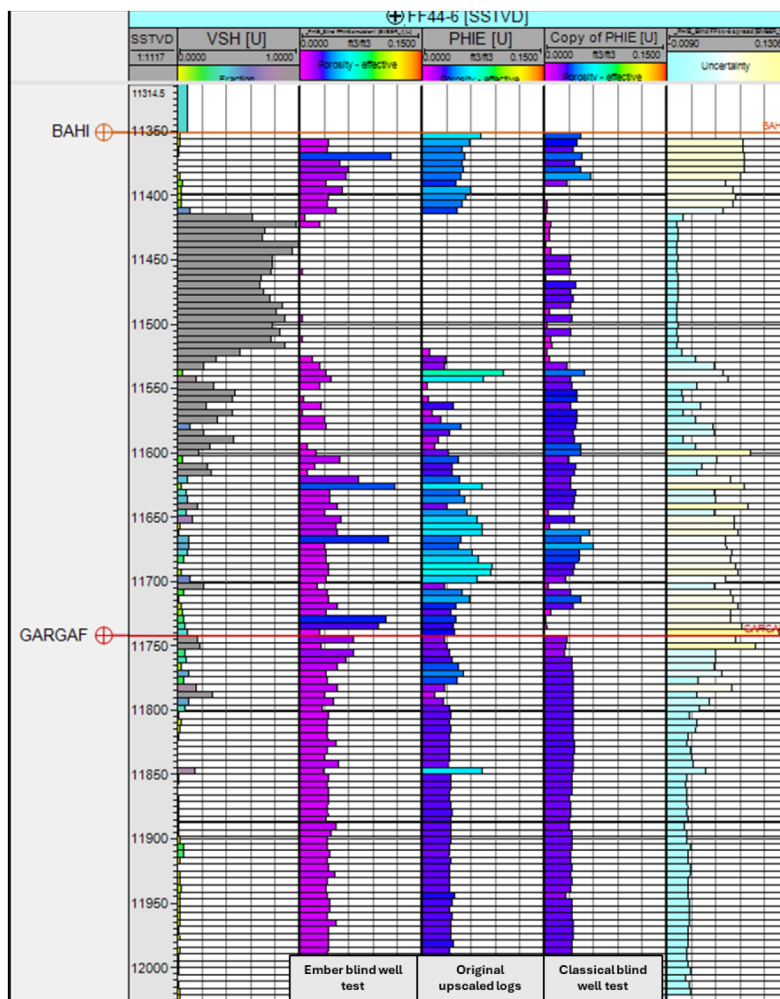


Figure 4—Comparison of Ember Vs Geostatistical Methods Blind Well Test Vs Original log

A key outcome is the notable improvement in model accuracy when using the EMBER approach, as evidenced by the correlation coefficient, a statistical measure that reflects how well the predicted porosity values match the actual, measured values from well logs. In this case, the correlation coefficient increased significantly from 0.32 using the classical geostatistical method to 0.60 with EMBER. This represents an 87.5% improvement in predictive performance, demonstrating that EMBER captures the underlying relationships in the data more effectively.

This enhanced correlation translates into more reliable porosity models, especially in interwall areas where direct data is unavailable. As a result, reservoir engineers and geoscientists can have greater confidence in using EMBER-based models for volumetric estimation. The improvement in correlation also

highlights EMBER’s ability to generalize from limited well data while leveraging broader geological and geophysical context.

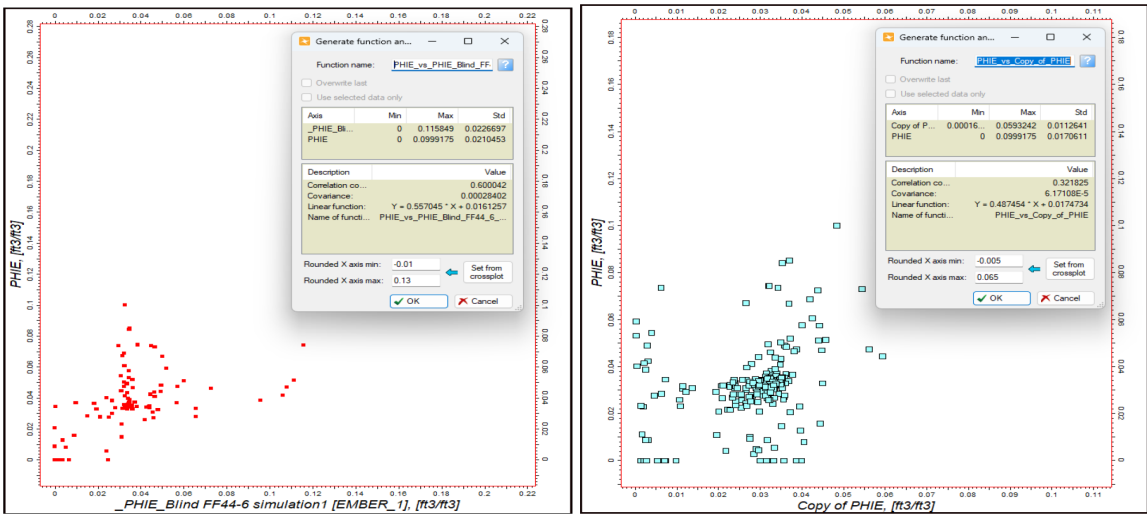


Figure 5—Cross-plots illustrate the comparison between Ember and the classical geostatistical method

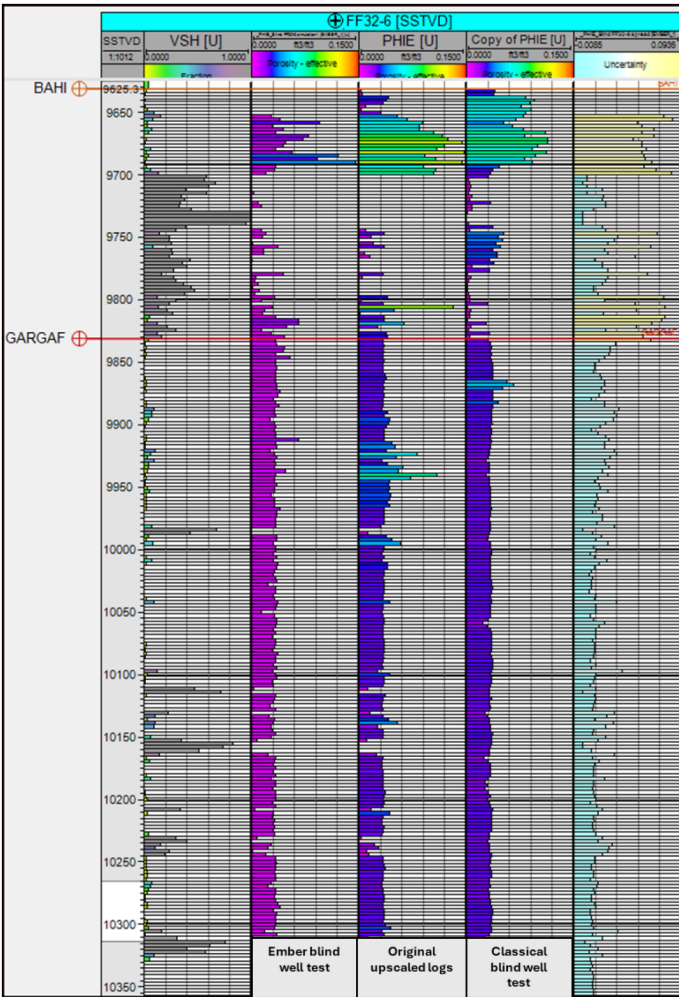


Figure 6—An additional example comparing the Ember blind well test to the classical blind well testing approach

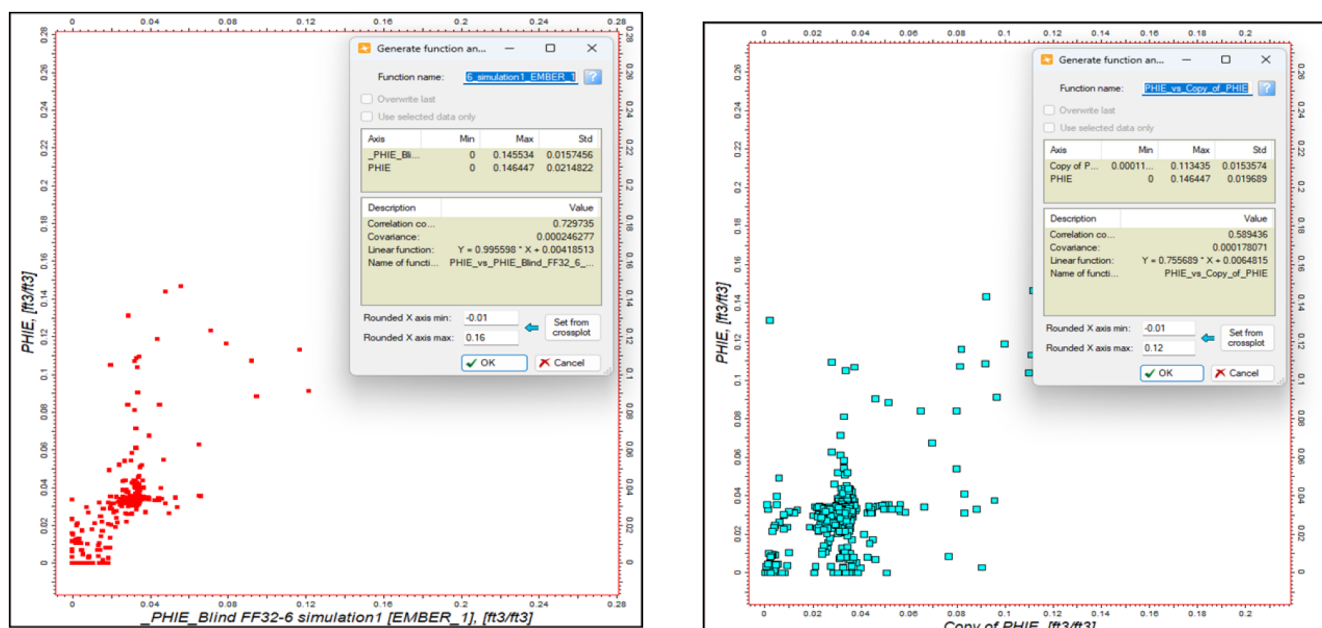


Figure 7—The correlation coefficient is 0.72 when applying Ember, while 0.59 when using the classical method

### Reservoir Modeling Approach and Results: Upper Sabal Reservoir 103 D(Classical Method)

There are two sources of porosity information: petrophysical evaluation of well logs and transformed porosity from inverted seismic acoustic impedance. To fully integrate those porosity information sources, porosity was geo-statistically populated well porosity using Sequential Gaussian Simulation but using the seismic-driven porosity as a secondary trend (i.e., local variable statistical mean) and conditioned to a previously simulated E-facies model (i.e., porosity distribution will be biased by E-facies ).

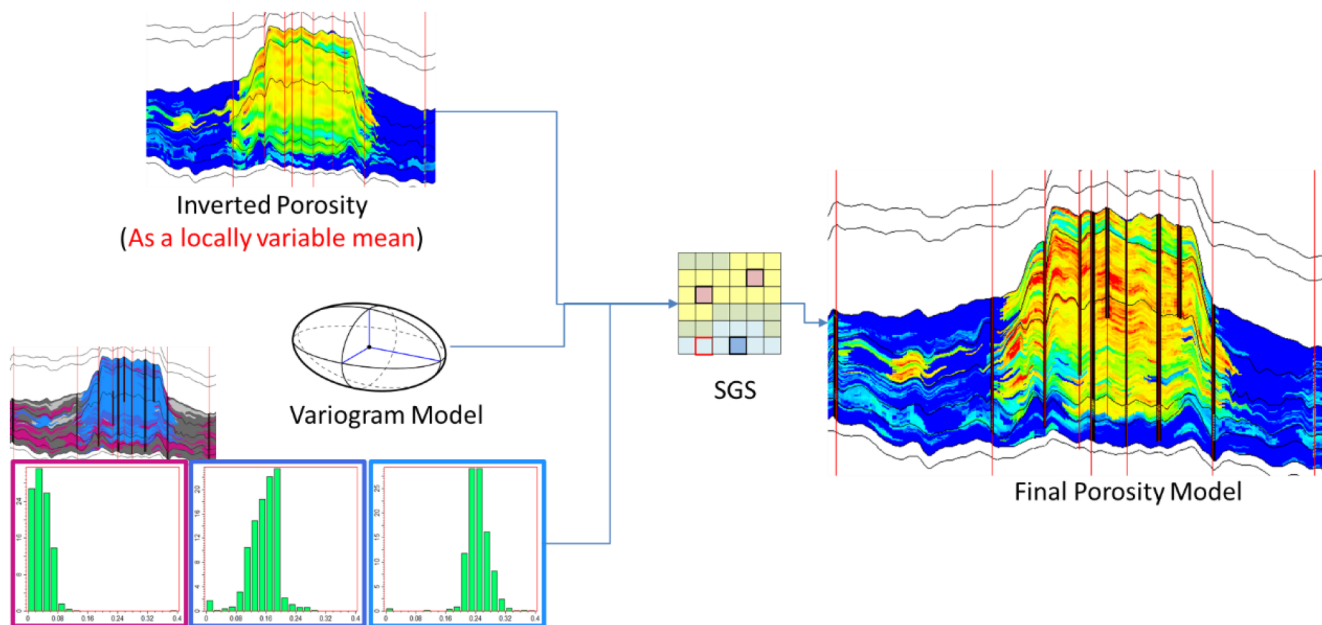


Figure 8—Classical porosity modeling workflow.

### Seismic-Driven Porosity

Theoretically, acoustic impedance is inversely proportional to porosity. That was checked with the available well data, and the correlation coefficient is almost -0.9 as shown below

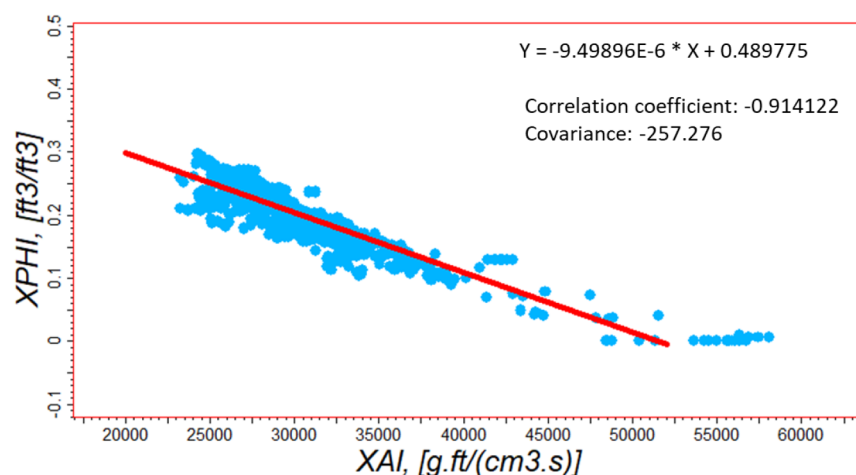


Figure 9—Porosity vs. acoustic impedance cross plot from well data.

The well data suggest that rocks with AI greater than 52000 ft/s.g/cc have zero porosities. Due to the fact that argillaceous limestone has almost the same acoustic impedance range of good-quality limestones, then the porosity-acoustic-impedance transform will be applied only on the Bayesian-classified limestones whose acoustic impedance is below 52000 ft/s.g/cc, and setting anything else to zero porosity.

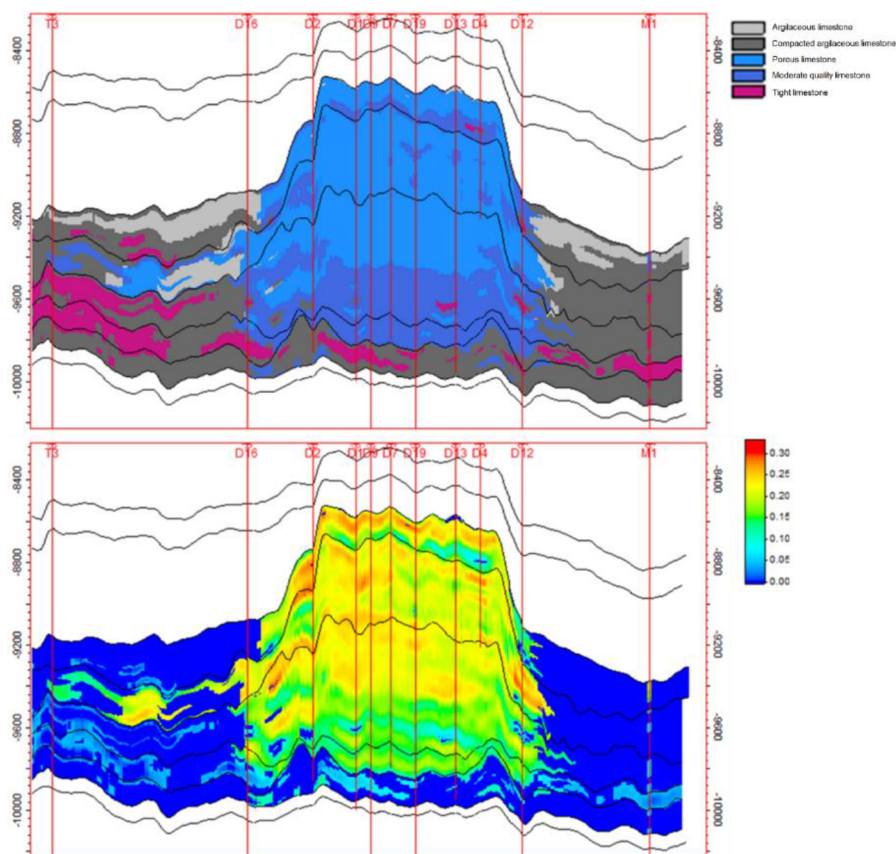


Figure 10—cross-section for facies model, model, and AI-transformed porosity

Comparison of the seismic-driven porosity with the actual porosity at the well locations suggests that the prediction is reasonable to very good (Figure 11)

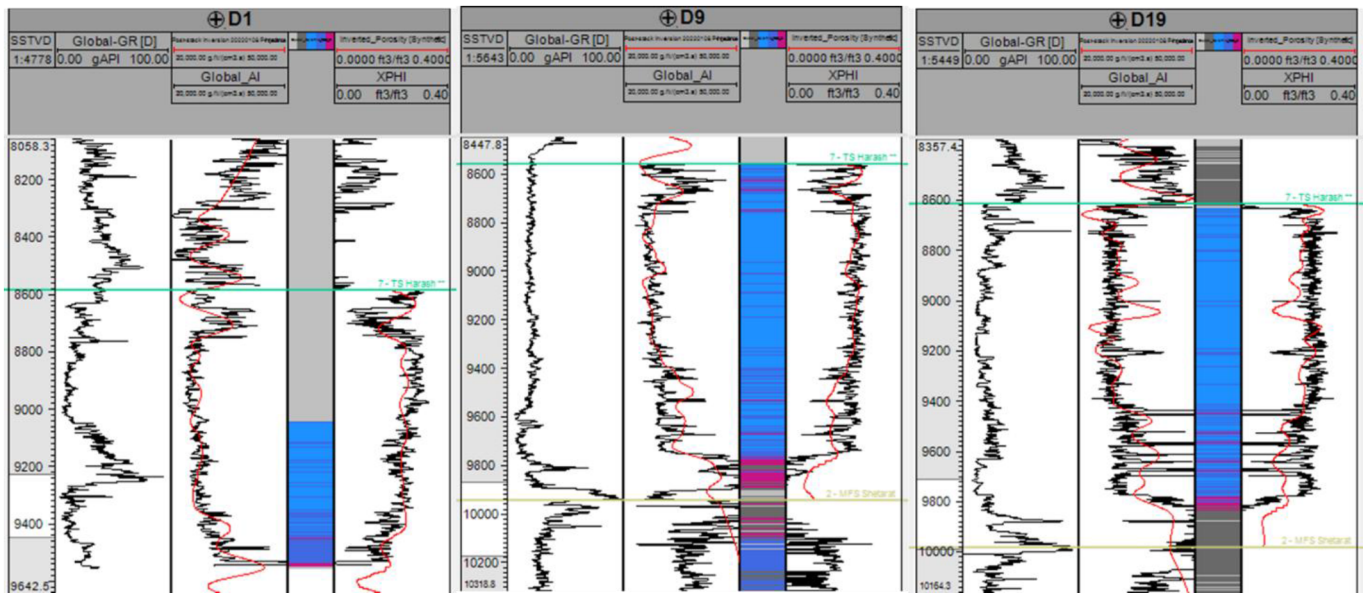


Figure 11—QC Example of the seismic-driven porosity in red and actual porosity in black in the right track.

Porosity simulation was done using SGS, which was fed by the above E-facies distributions (E-facies model), geostatistical model, and the seismic-driven porosity as a co-kriging variable. The result porosity simulation considering all input data

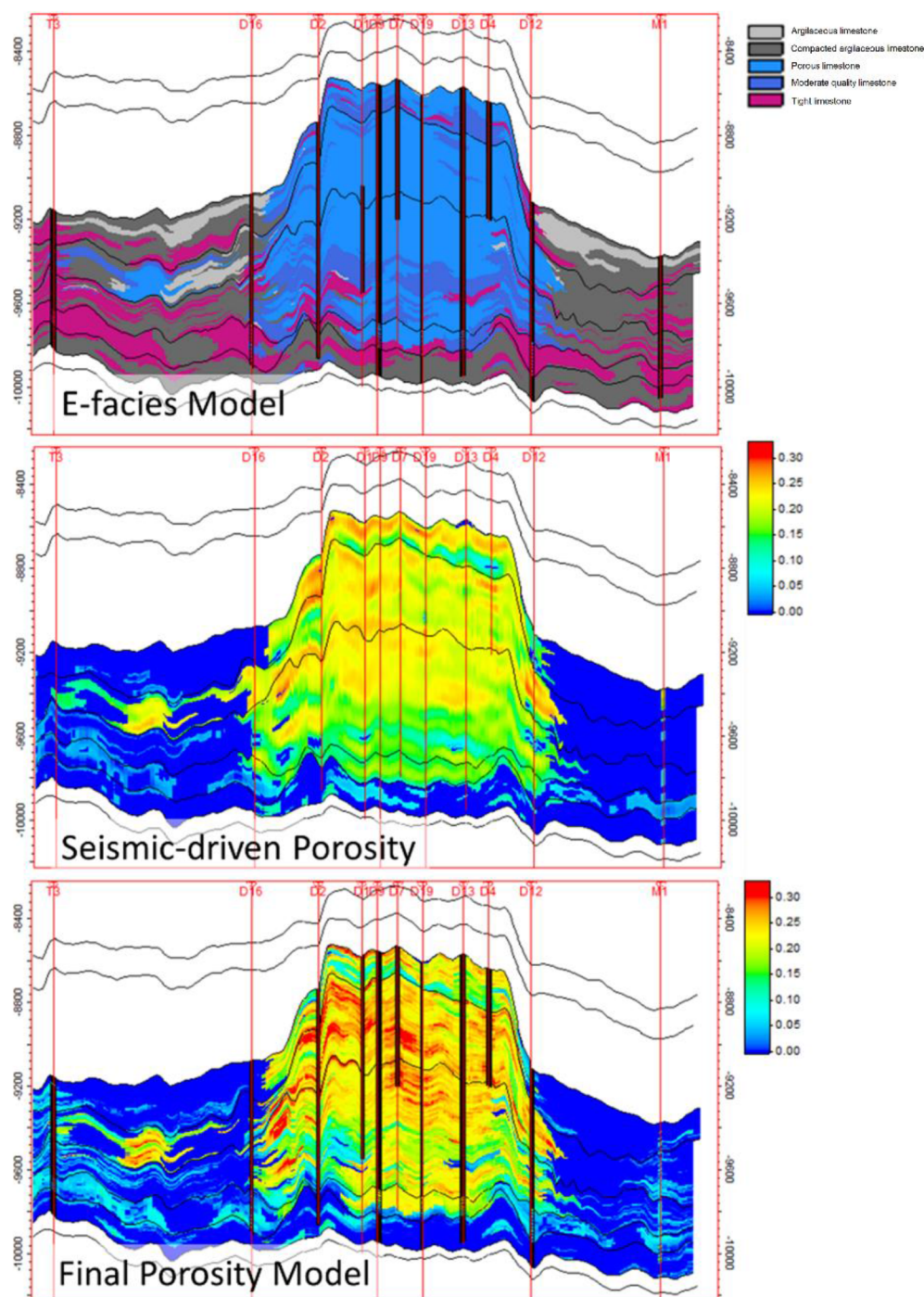


Figure 12—Cross-section illustrating facies distribution, seismic-derived porosity, and the final porosity model generated using Sequential Gaussian Simulation (SGS)

### Reservoir Modeling Approach and Results: Upper Sabal Reservoir 103 D(EMBER Method)

During the implementation of Ember, we used the same input used to create the porosity (SGS algorithm), which allowed us to compare the results with the classical approach results more accurately. Below is the Ember process window.

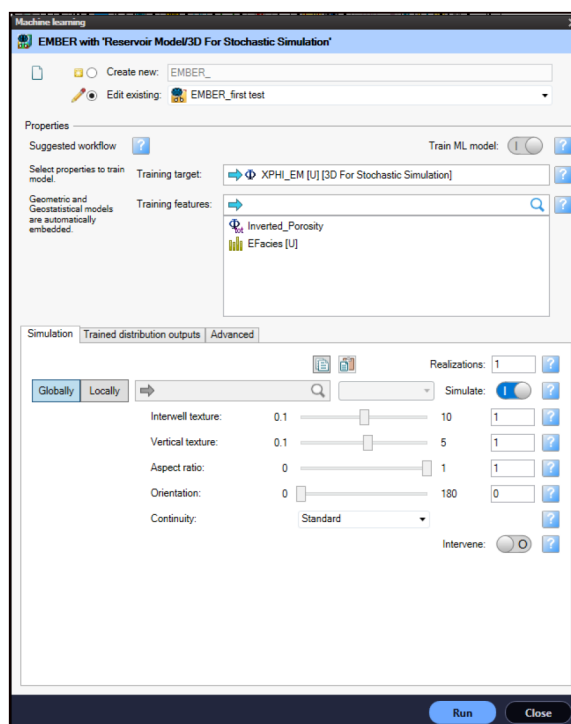


Figure 13—Ember model dialog

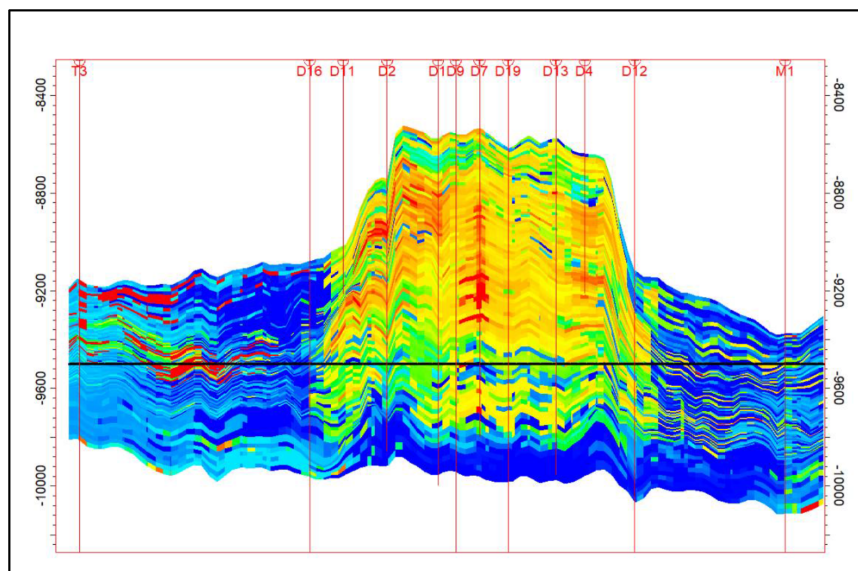


Figure 14—Section illustrates the porosity model using Ember, where the highest porosity is concentrated in the reef buildup

Overall, the porosity results generated by EMBER are comparable to those produced using the Sequential Gaussian Simulation (SGS) method. However, some differences are observed at the edges of the model. In the SGS workflow, engineers applied a geometric constraint to manually reduce porosity values near the model boundaries. In contrast, EMBER, being a data-driven approach, did not incorporate this constraint. As a result, EMBER predicted slightly higher porosity values at the model edges compared to the SGS model. Below is the blind well test result for one of the wells in the model, used to evaluate the accuracy of the porosity predictions.

Figure 15 shows a comparative blind well test for porosity prediction at Well D9, evaluating the performance of the EMBER machine learning algorithm versus the classical geostatistical approach.

The logs are divided into four tracks: predicted porosity from EMBER (left), original upscaled log porosity (center-left), classical geostatistical blind test (center-right), and associated uncertainty (right). The stratigraphic column highlights the main reservoir intervals: Upper Sabil formation.

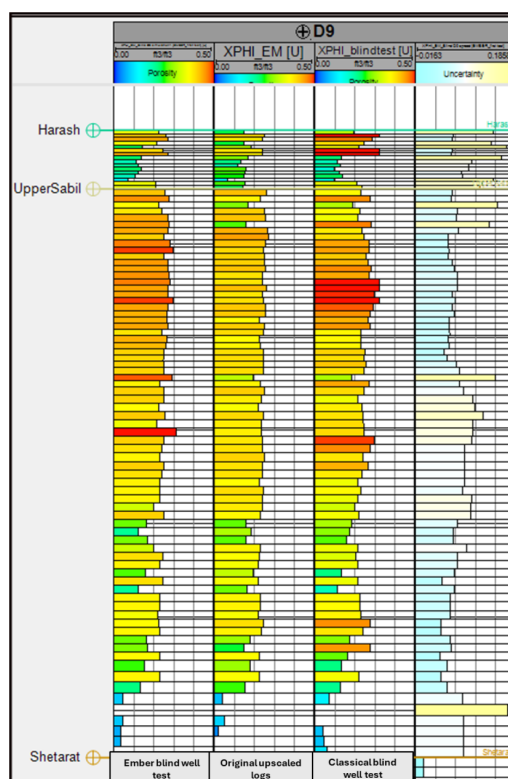


Figure 15—Very good agreement for both classical and Ember, However, Ember is slightly better in predicting the upper and the middle section for this well

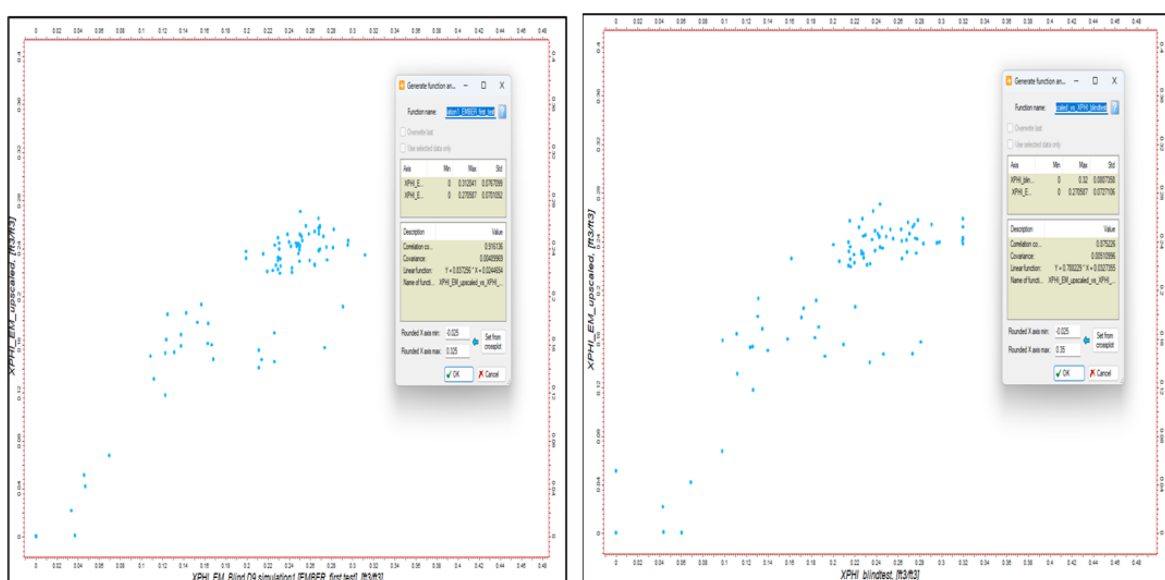


Figure 16—Correlation coefficient is 0.91 for Ember and 8 for the classical methods

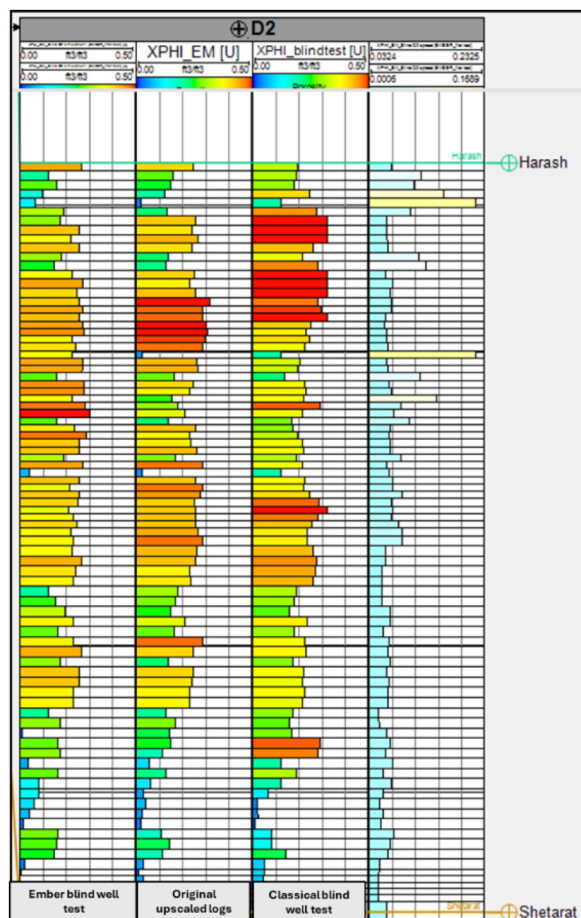


Figure 17—Another example comparing the blind well test using the Ember approach versus the geostatistical approach

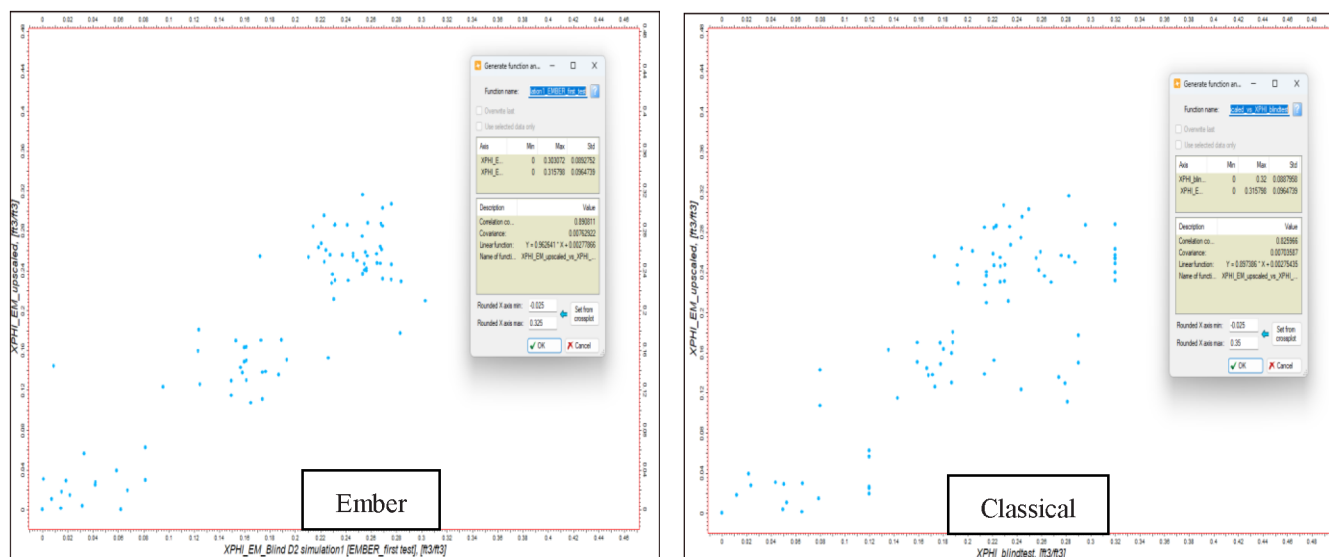


Figure 18—The correlation coefficient is 0.89 for Ember and 0.81 for the classical methods

From the EMBER blind test results (first track), we observe that the predicted porosity closely mirrors the actual upscaled porosity log, especially within the Upper Sabal Formation, which is the primary reservoir unit. The color continuity and alignment with the original log suggest that EMBER accurately captures

porosity distribution and reservoir heterogeneity. Notably, it performs well across both high and low porosity intervals, identifying vertical trends and transitions without over-smoothing or introducing noise.

In contrast, the classical geostatistical blind test (third track) tends to underestimate porosity in the same intervals. There is a visible mismatch, particularly in the higher porosity zones of the Upper Sabil and Harash formations, where classical methods show more abrupt transitions and fail to capture fine-scale variability.

The uncertainty panel further supports EMBER's strength, as lower uncertainty is observed where EMBER predictions align well with original log data. This suggests higher confidence in the machine learning-based model.

Overall, the figure illustrates how EMBER delivers a more geologically consistent and statistically accurate porosity model, outperforming traditional methods.

## Conclusions

This study demonstrates the effectiveness of the EMBER machine learning algorithm as a modern alternative to traditional geostatistical methods for porosity modeling in geologically complex reservoirs. While classical approaches such as Sequential Gaussian Simulation (SGS) have been widely used in reservoir modeling, they often involve complex, time-consuming workflows and rely heavily on user-defined parameters like variograms. In contrast, EMBER simplifies the process by using a data-driven approach to automatically learn the relationships between geological and geophysical inputs and the target reservoir properties.

In two case studies from the Sirte Basin, EMBER delivered porosity models with accuracy equal to or better than classical methods. Notably, the correlation coefficient improved significantly, from 0.32 using SGS to 0.60 with EMBER, highlighting its superior ability to capture underlying geological patterns. Additionally, EMBER reduced modeling time by up to 70%, streamlining workflows and enabling faster decision-making in field development planning.

The results also showed that EMBER could generalize well in areas with sparse data, though its performance at model boundaries may differ due to the lack of geometric constraints typically applied in classical workflows. Despite this, its flexibility, efficiency, and predictive accuracy make it a powerful tool for modern reservoir characterization.

Moreover, EMBER's integration of multiple data types - well logs, seismic attributes, and facies - demonstrates its robustness and adaptability in complex geological settings. The reduced dependency on manual calibration steps enhances consistency and minimizes human bias, making the modeling process more objective and reproducible across teams and projects.

Looking ahead, the use of EMBER and similar machine learning tools is expected to expand in subsurface workflows, particularly as more high-resolution geological and geophysical data become available. Future developments could include hybrid approaches that combine EMBER's machine learning strengths with the spatial control of geostatistics, further improving model realism and reservoir predictability.

In summary, EMBER provides a practical, efficient, and scalable solution for porosity modeling, helping bridge the gap between data science and geological understanding in reservoir management.

## Acknowledgments

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## Nomenclature

**Variogram** A function describing spatial continuity and correlation of a variable, used to guide interpolation in geostatistical simulations.

- Sequential Gaussian Simulation (SGS)** A pixel-based geostatistical algorithm used to generate multiple realizations of a petrophysical property based on a Gaussian distribution.
- Acoustic Impedance (AI)** The product of rock density and seismic velocity, often used as a proxy for porosity in seismic inversion.
- E-facies** logs derived facies classifications.
- EMBER** A machine learning-based modeling tool that integrates geological, petrophysical, and seismic data to predict reservoir properties.

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